

Part A. Data

Source: World Bank, World Development Indicators — series "Life expectancy at birth, total (years)"

URL: <https://data.worldbank.org/indicator/SP.DYN.LE00.IN?locations=PH>

Description: The dataset reports the average number of years a newborn in the Philippines is expected to live if current mortality patterns stay the same, for every year from 1990 to 2023.

Independent variable x: Year

Dependent variable y: Life expectancy at birth (years).

Table of observations

Year	Life Exp. (yrs)	Year	Life Exp. (yrs)
1990	64.382	2007	68.549
1991	64.622	2008	68.643
1992	65.353	2009	68.631
1993	65.661	2010	68.883
1994	66.000	2011	69.096
1995	66.311	2012	69.237
1996	66.691	2013	69.283
1997	66.994	2014	69.311
1998	67.212	2015	69.450
1999	67.503	2016	69.490
2000	67.789	2017	69.966
2001	67.829	2018	69.761
2002	67.886	2019	69.680
2003	68.049	2020	70.097
2004	68.301	2021	66.675
2005	68.423	2022	69.472
2006	68.172	2023	69.833

Part B. Python Script

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression

# 1. Data
data = {
    'Year': [1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999,
             2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009,
             2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019,
             2020, 2021, 2022, 2023],
    'LifeExpectancy': [64.382, 64.622, 65.353, 65.661, 66.000, 66.311, 66.691, 66.994, 67.212,
                      67.503,
                      67.789, 67.829, 67.886, 68.049, 68.301, 68.423, 68.172, 68.549,
                      68.643, 68.631,
                      68.883, 69.096, 69.237, 69.283, 69.311, 69.450, 69.490, 69.966,
                      69.761, 69.680,
                      70.097, 66.675, 69.472, 69.833]
}

df = pd.DataFrame(data)

# 2. Simple linear regression
def simple_linear_regression(X, y):
    model = LinearRegression()
    model.fit(X, y)
    print("Simple Linear Regression Coefficients:")
    print("Slope (Coefficient):", model.coef_[0])
    print("Intercept:", model.intercept_)
    print(f"Y = {model.intercept_:.4f} + {model.coef_[0]:.4f}X")
    print("-----")
    return model

X = df[['Year']]
y = df['LifeExpectancy']

model = simple_linear_regression(X, y)
a1 = model.coef_[0]          # slope
a0 = model.intercept_        # intercept

y_pred = model.predict(X)

# 3. Goodness-of-fit statistics: Sr (SSE), r^2, standard error sy/x
n = len(df)
residuals = y.values - y_pred

Sr = np.sum(residuals ** 2)          # sum of squares of residuals (SSE)
St = np.sum((y.values - y.mean()) ** 2) # total sum of squares
r2 = 1 - (Sr / St)                   # coefficient of determination
syx = np.sqrt(Sr / (n - 2))          # standard error of the estimate

print(f"Sr (SSE)           = {Sr:.4f}")
print(f"r^2               = {r2:.4f}")
print(f"Standard error (sy/x) = {syx:.4f}")
```

```
print("-----")
```

```
# 4. Prediction for a year not in the dataset
```

```
year_new = 2027
```

```
pred_new = model.predict(pd.DataFrame({'Year': [year_new]}))[0]
```

```
print(f"Predicted life expectancy for {year_new}: {pred_new:,.2f} years")
```

```
print(f"Y = {a0:,.4f} + {a1:,.4f}({year_new}) = {pred_new:,.4f}")
```

```
print("-----")
```

```
# 5. Plot: data points with fitted line
```

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(df['Year'], df['LifeExpectancy'], color='#1f4e79', label='Observed data')
```

```
plt.plot(df['Year'], y_pred, color='#c8963e', linewidth=2, label='Fitted line')
```

```
plt.xlabel('Year')
```

```
plt.ylabel('Life expectancy at birth (years)')
```

```
plt.title('Philippines Life Expectancy vs. Year (1990-2023)')
```

```
plt.legend()
```

```
plt.tight_layout()
```

```
plt.savefig('fit_plot.png', dpi=150)
```

```
plt.show()
```

```
# 6. Residual Plot
```

```
plt.figure(figsize=(8, 5))
```

```
plt.scatter(df['Year'], residuals, color='#1f4e79')
```

```
plt.axhline(0, color='#c8963e', linewidth=2)
```

```
plt.xlabel('Year')
```

```
plt.ylabel('Residual (years)')
```

```
plt.title('Residual Plot')
```

```
plt.tight_layout()
```

```
plt.savefig('residual_plot.png', dpi=150)
```

```
plt.show()
```

Part C. Regression Results

Regression equation

$$Y = -206.4612 + 0.1368 X$$

Slope (a_1): 0.1368 — years of life expectancy gained per additional calendar year

Intercept (a_0): -206.4612

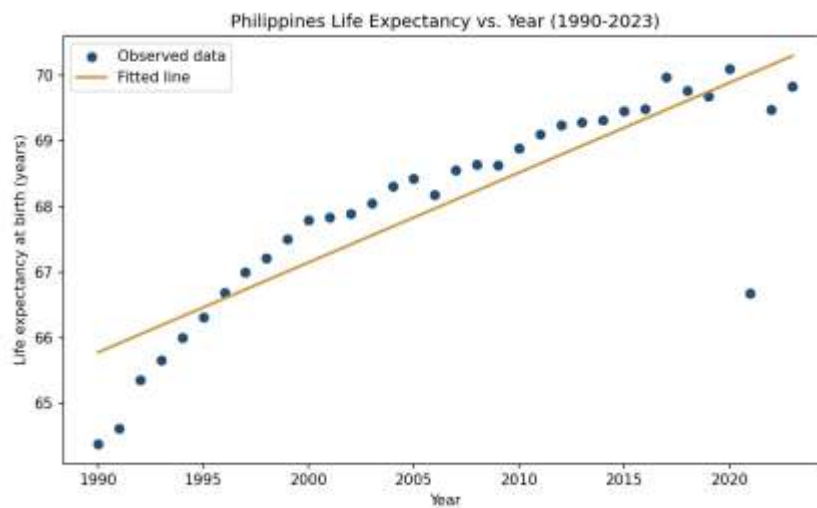
Goodness of fit

Sr (SSE): 20.3018

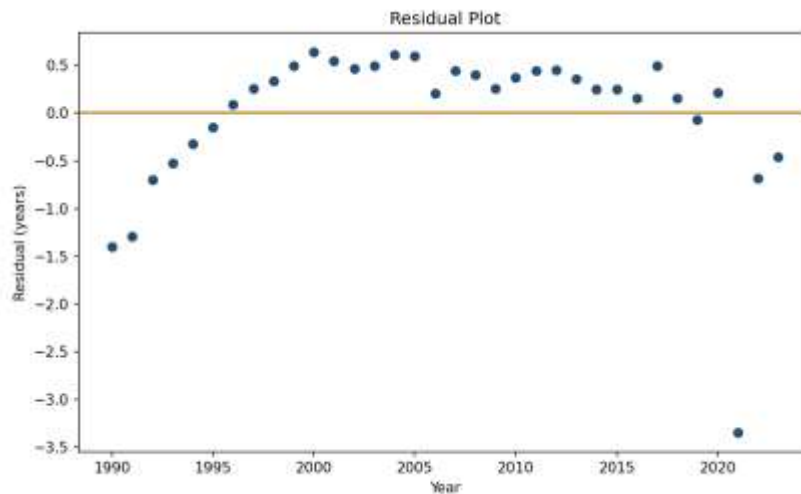
r^2 : 0.7510

Standard error (s_y/x): 0.7965 years

Fitted line over the data



Residual plot



Prediction

For $x = 2027$ (a year not in the dataset):

$$Y = -206.4612 + 0.1368(2027) = 70.84 \text{ years}$$

The model predicts a Filipino born in 2027 would have a life expectancy at birth of about 70.8 years, consistent with the gradual upward trend in the data.

Interpretation

Slope: On average, life expectancy at birth in the Philippines increased by about 0.137 years for every additional calendar year between 1990 and 2023 — roughly 1.4 years per decade.

Intercept: The intercept (-206.46) has no direct physical meaning here since it corresponds to year 0; it simply positions the line correctly over the 1990-2023 range.

Goodness of fit: $r^2 = 0.751$ means about 75% of the year-to-year variation in life expectancy is explained by the linear trend with year, while the remaining 25% is due to other factors (economic shocks, health-policy changes, epidemics, etc.). The standard error of the estimate, 0.80 years, indicates that individual yearly values typically deviate from the fitted line by less than one year.

Residuals: Most residuals are small and scattered closely around zero with no strong pattern, supporting a linear trend as a reasonable model. The clear exception is 2021, which shows a large negative residual: the COVID-19 pandemic caused a sharp, temporary drop in life expectancy that the straight-line model cannot capture, since a single global shock is not part of a steady long-term trend.